

Coding & Solution

Date	29 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements. Financial Health Ratios stress assessment; Prioritized loan lists sorting; Google Gemini API negotiation emails hardship letters generation; SQL history logging.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	Github Link / Website Link
Programming Language(s)	Python 3.10+ (Backend), JavaScript/JSX (Frontend)
Framework(s) Used	FastAPI (Backend), React.js + Vite (Frontend), SQLAlchemy (ORM)

Key Features Implemented	• JWT user authentication with bcrypt password hashing • Loan CRUD with intelligent priority scoring (overdue severity, interest rate, outstanding amount) • Financial health calculator (EMI ratio, debt-to-income ratio, stress level, health score) • Settlement prediction engine with recommended offer percentages • AI negotiation letter generator via Google Gemini API (3 tones) • Automatic fallback to rule-based template letters • Real-time dashboard with aggregated financial statistics • Protected React SPA with sidebar navigation and Recharts visualizations
Pending / Incomplete Features	Cloud deployment, multi-language support, mobile app, PDF download of letters, real-time notification system
Setup / Run Instructions	1. Clone repo → 2. Backend: cd backend && pip install -r requirements.txt → 3. Create .env with GEMINI_API_KEY → 4. uvicorn main:app --reload → 5. Frontend: cd frontend && npm install → 6. npm run dev → 7. Open http://localhost:5173



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

